

FerroDB • The core, connected

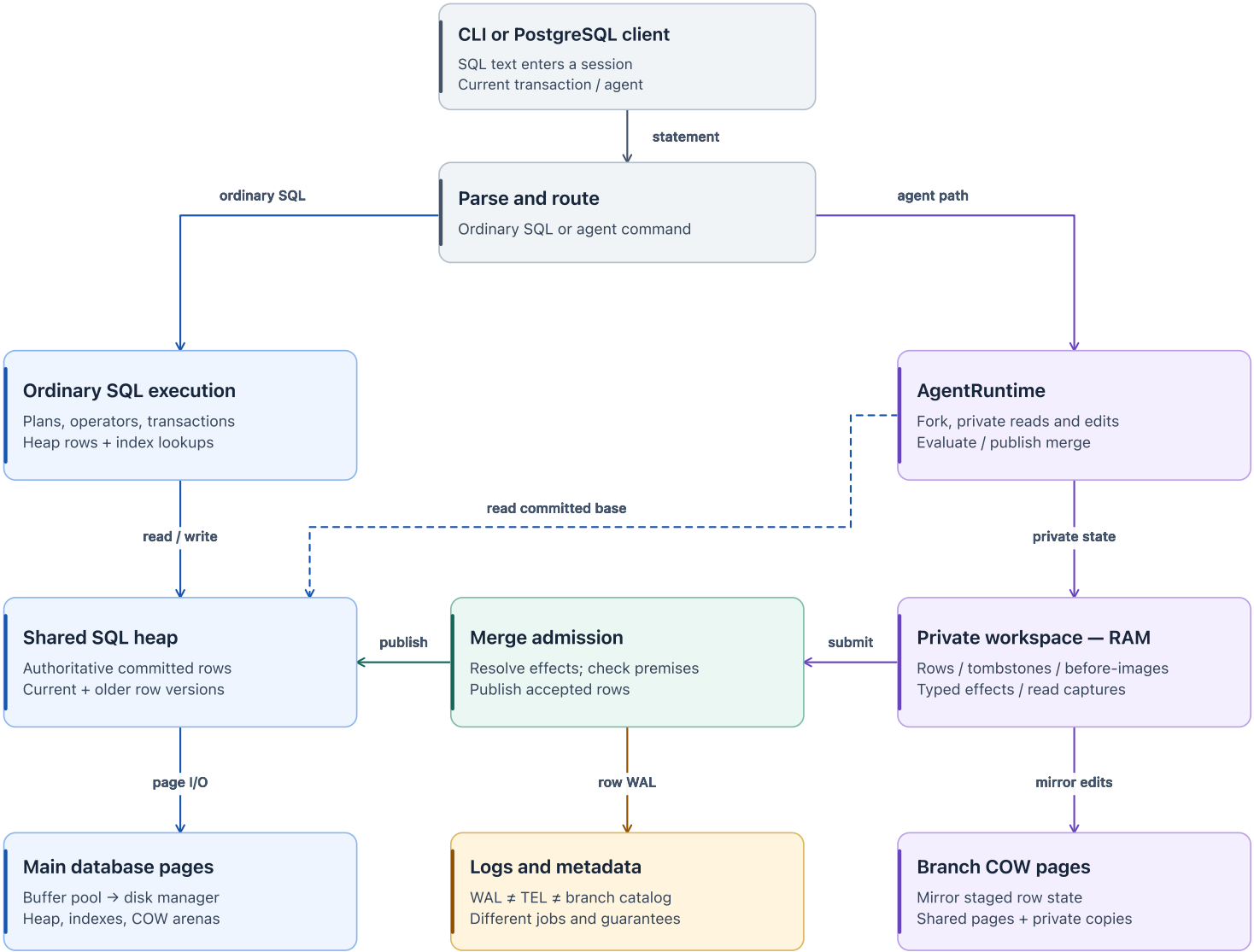
An implementation map, from SQL to bytes and back. Read the numbered views in order; select a box for its mechanism and evidence. Solid arrows show the labeled dependency or data flow. Dashed arrows mark a separate path, optional component, or later action as labeled.

Revision `fc9556a742f6` 2026-09-23

- Ordinary SQL / shared data
- Private branch / COW
- Log / durability
- Validation / boundary
- Ownership / cleanup
- Entry point / separate component

The system, in one view

Start here. Follow the blue SQL path or the purple agent path; they meet when a merge publishes rows.



Ordinary DML and accepted merge rows both use WAL; private COW staging has separate durability.

The decisive distinction

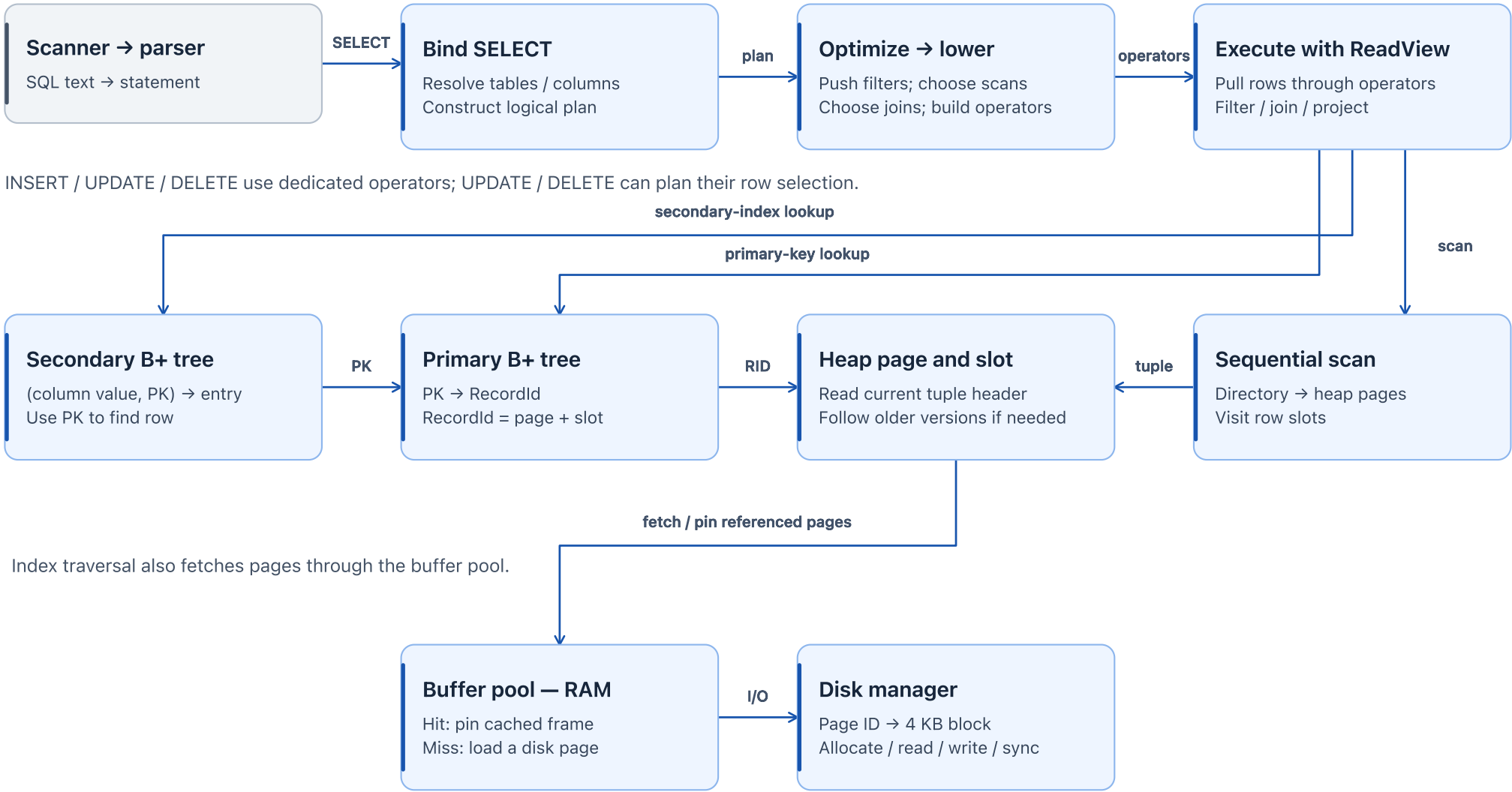
SQL row versions, private workspace state, and COW page versions are different structures. The detailed views below follow each separately.

Actual server concurrency

Eligible reads can run together. Current pgwire statements that write take an exclusive catalog gate and drain readers; multiple agents do not imply simultaneously executing SQL writers.

From SQL to a physical row

This is the ordinary SQL path. Primary and secondary indexes lead to heap row locations, not branch roots.



Two kinds of synchronization

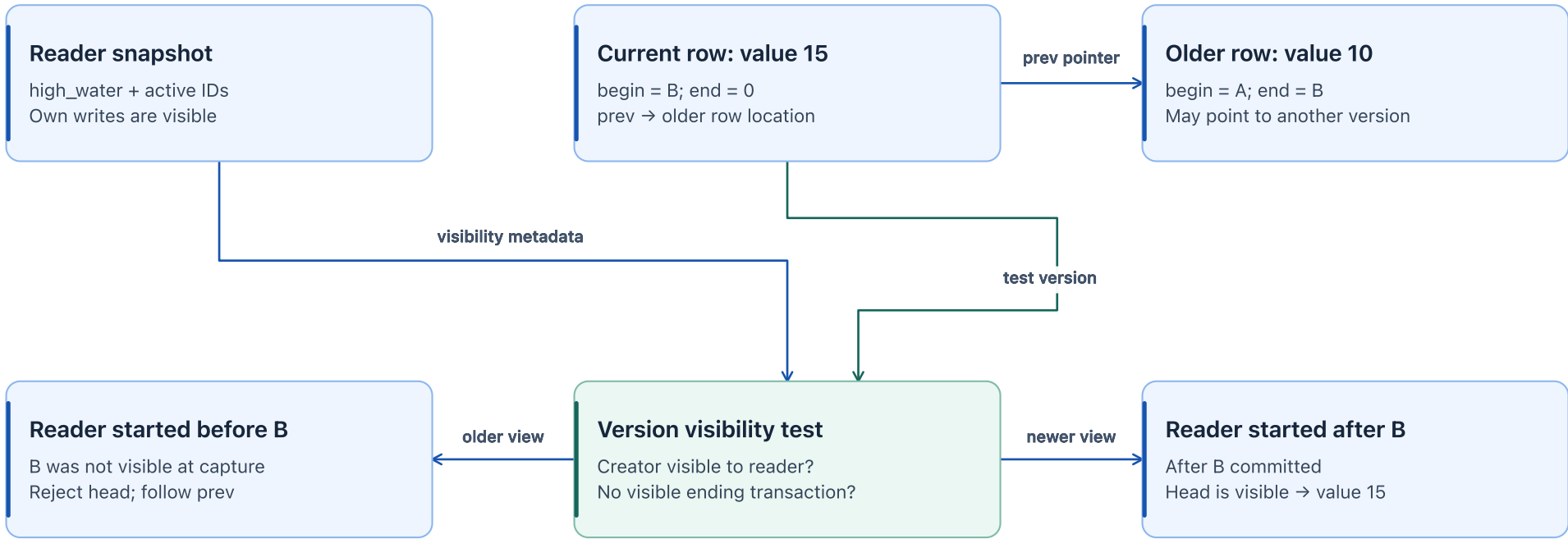
MVCC decides logical visibility. Page latches protect physical tree structure during traversal/splits. Buffer pinning protects residency. None of these substitutes for the others.

Index maintenance is distinct

Ordinary indexes mutate in place. Deletes retain entries and scans filter stale candidates. Heap recovery is followed by index rebuilding in the CLI; this is not per-index-operation WAL redo.

MVCC: time belongs to the reader

A snapshot is a visibility rule over transaction IDs. It is not a copied database and not a change log.



UPDATE: preserve old tuple → write new head. DELETE: mark end transaction; keep history.

Overlapping writes can raise a write conflict. MVCC visibility alone does not prove serializable execution.

A snapshot does not lock in agent SQL

This diagram describes ordinary transaction snapshots. Agent SQL obtains fresh committed base rows for each read and overlays private state; it does not retain this full view from branch fork.

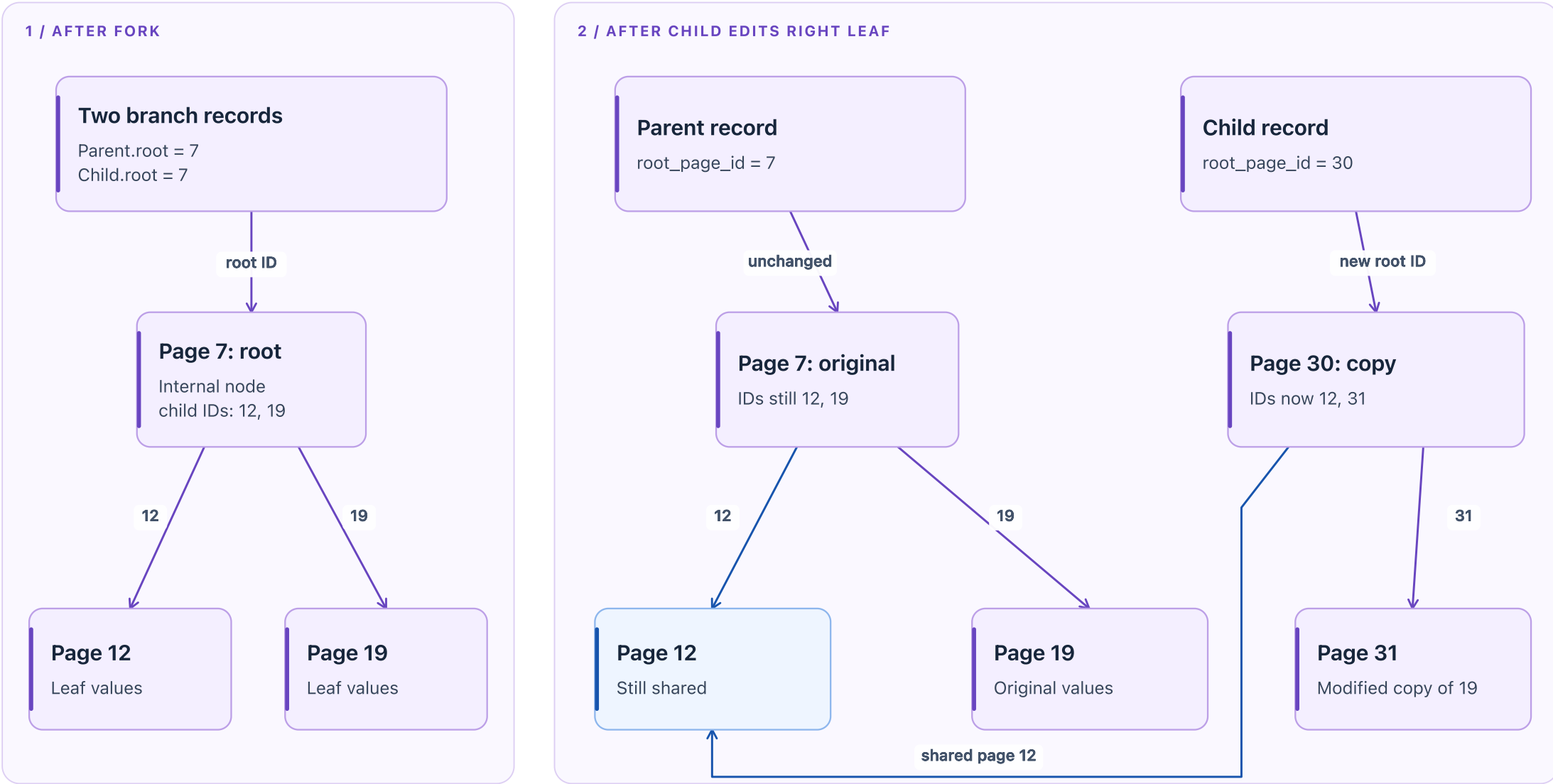
History has a separate lifetime

The branch reaper frees branch COW pages. It is not a vacuum for old heap row versions; no general MVCC vacuum is represented here.

Ordinary operation	Row/version action	Index action
INSERT	Create tuple with begin=my transaction	Primary upsert; secondary entries
UPDATE	Preserve old copy; new head points to it	Maintain mapping if location/indexed value changes
DELETE	Set ending transaction; retain historical row	Entries can remain; reads check visibility

Fork and shadow paging, with real pointer roles

Illustrative IDs. This is the branch COW tree, not a frozen copy of every ordinary SQL table.



Assume shared starting pages; no split or boundary repair. Copy 19 and the route to it; keep 12 shared.

RAM shares a second structure

Fork also clones roots of immutable workspace maps. These are balanced binary trees in RAM, not disk B+ tree pages. Map edits copy their path; later parent-private edits stay isolated.

Exclusively owned pages can change in place

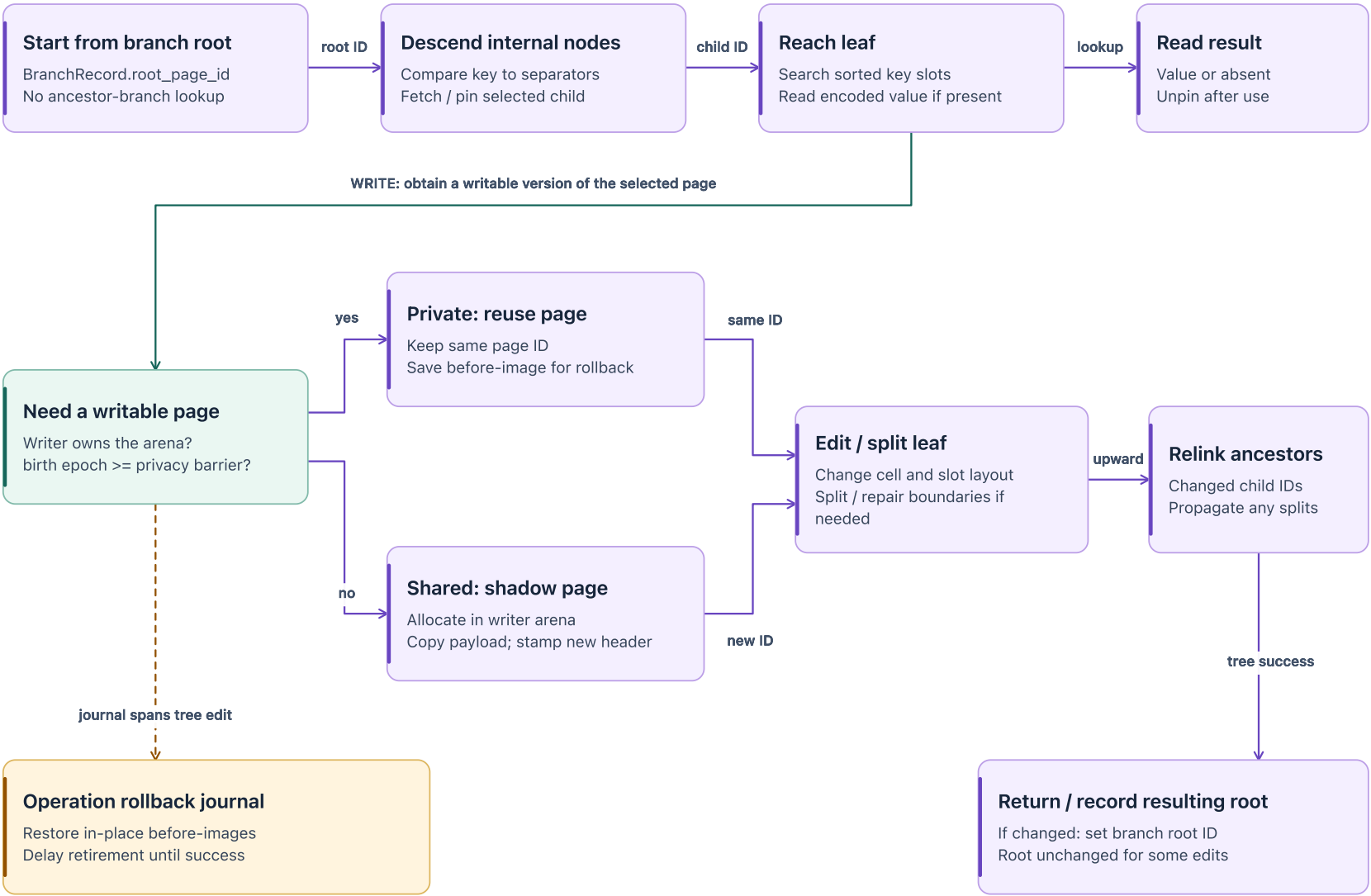
The writer must own the page and its birth epoch must meet the privacy barrier, which includes the latest live child fork. Splits or boundary repairs can allocate extra pages.

Avoiding a full data copy is not zero cost

Fork still performs catalog, synchronization and durability work. Branches also add effect capture, workspace and page-management costs; no cheaper-than-transactions claim is made.

The COW engine: lookup, ownership, edit, relink

The tree algorithm operates on page IDs. Arena ownership decides whether a page can change in place.



Bounded copying is the benefit

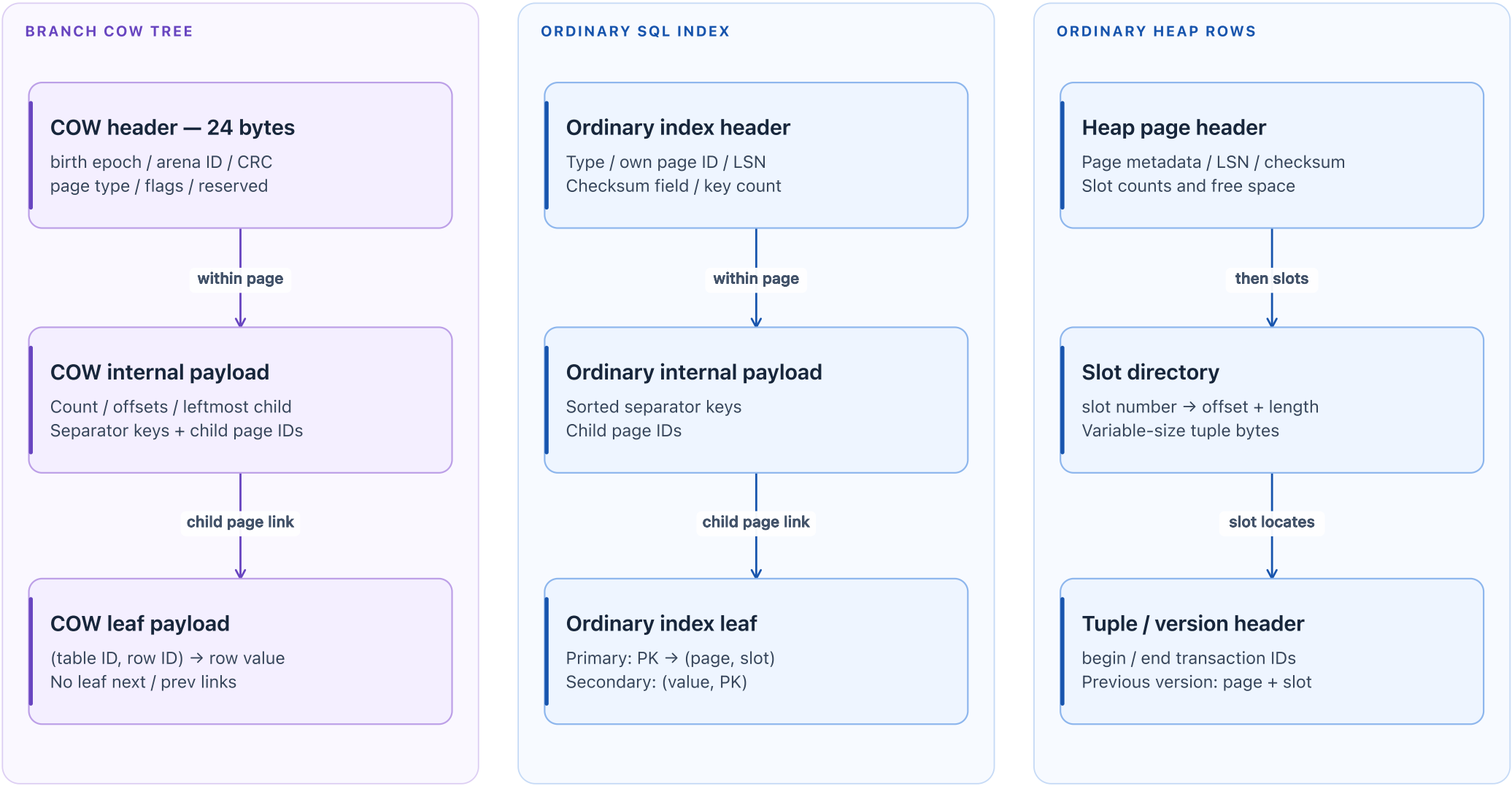
Unchanged subtrees remain shared. Read cost follows tree height, not branch ancestry depth. Actual write cost also includes allocations, splits, boundary repair and metadata/durability work.

Read and write structures are still distinct

This is the COW page engine. Ordinary SQL indexes have their own traversal/latching implementation; branch SQL reads still use current heap rows overlaid with a private RAM workspace.

Inside the pages: three layouts, three purposes

Page IDs locate blocks. Root IDs choose starting blocks. Child IDs route through trees. Row locations select heap slots.



A child link reaches a different page with its own header. Internal and leaf payloads are different node types.

Where is the root ID?

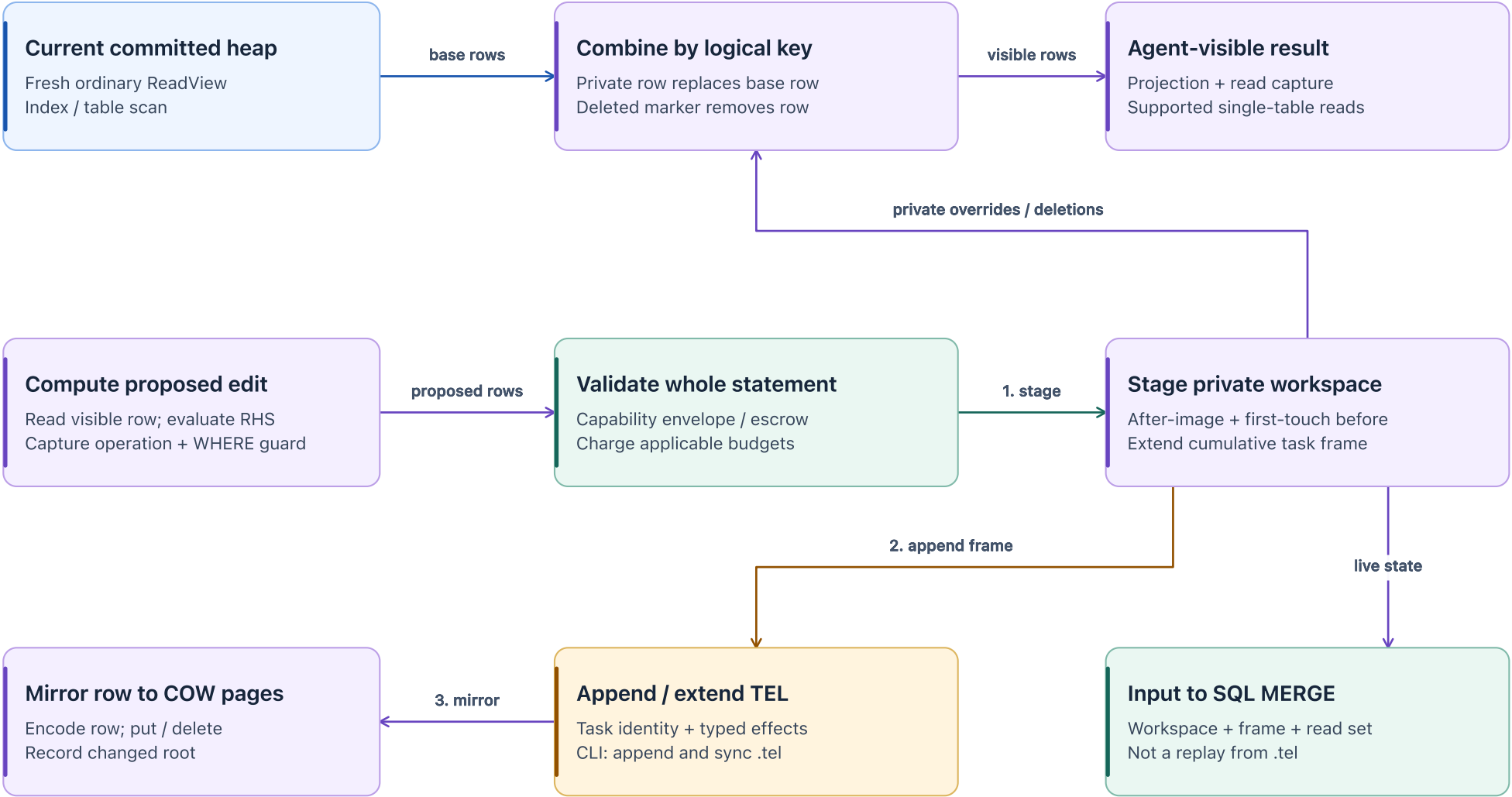
A branch record stores its COW root ID in the branch catalog. The catalog has its own separate tree root. Every page does not store the root of its tree.

Where is the page itself?

A buffer-pool hit reuses a RAM frame; a miss reads the requested page from disk. Fork does not load the whole tree. A page must be in memory before it can be edited.

What an agent actually reads and writes

Workspace state drives branch SQL. Its frame drives live SQL merge; the appended effect log is a separate representation.



A branch spans statements

A branch retains private proposals until merge or abandonment. Ordinary transaction IDs, effect-frame IDs, branch IDs and generations have different meanings.

Read validation is partial

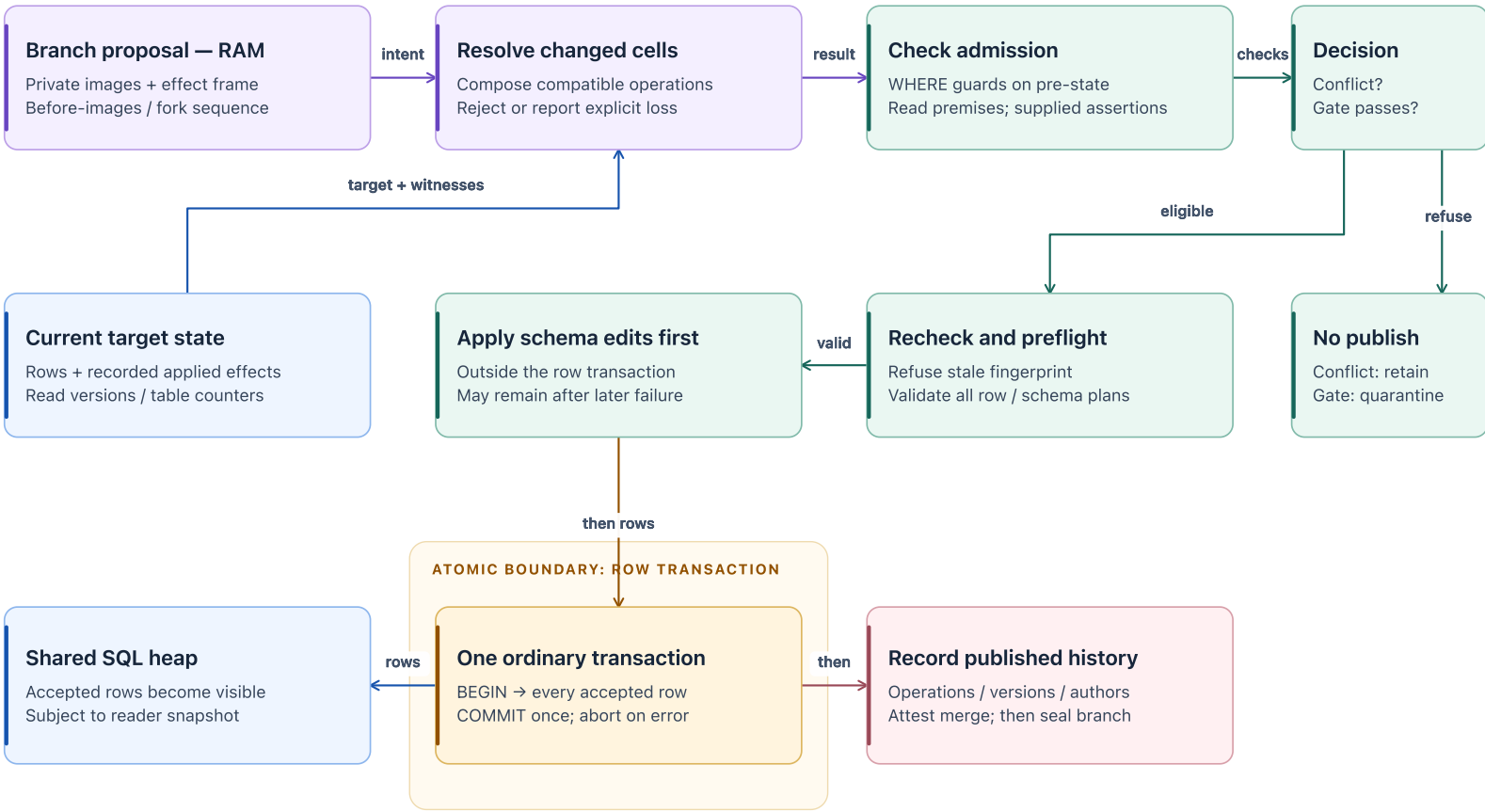
Exact point reads can detect later agent-published versions. Predicate summaries do not supply complete phantom validation; ordinary writes do not all update this same version map.

Three representations are not one commit

Workspace, TEL and COW mirror are staged in sequence. The per-tree rollback journal does not make the full sequence atomic or restore unfinished sessions after restart.

Merge means reconciliation, then publication

The live SQL merge uses operation intent, current target rows and checks. It does not replace the trunk COW root.



Arithmetic example

Start at qty 20. A captures Add(−5); B captures Add(−3). Merge A → 15; merge B applies −3 to current 15 → 12. Use point UPDATES without a prior stale explicit SELECT; additional checks can still refuse a merge.

Precondition versus postcondition

Two −12 withdrawals from 20: WHERE qty >= 12 rejects the second at current qty 8. WHERE qty >= 0 allows it and can produce −4. An explicit postcondition would test the proposed final value.

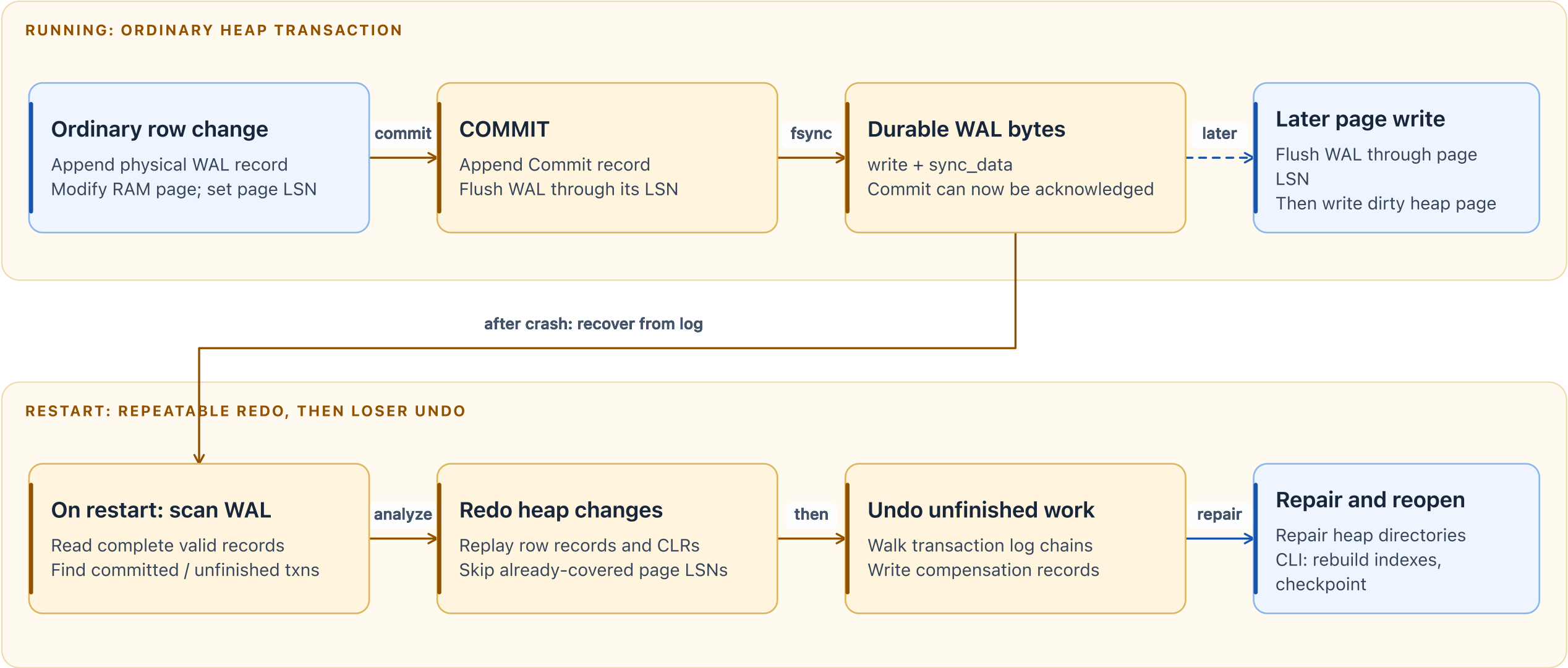
What is not atomic together

DDL, the row transaction, provenance/history recording and branch sealing are distinct stages. Frame retry deduplication does not establish exactly-once whole-merge retries after a crash.

Cell outcome	Meaning	Publication
Clean	No incompatible concurrent operation, or equal assignments	Eligible if checks pass
Commuting	Supported operations compose	Eligible if checks pass
Conflict	Incompatible operations / row conflict	No rows published
ResolvedWithLoss	Explicit incoming-wins policy discards a competing effect	Loss reported; checks still apply

WAL, commit, crash recovery

For ordinary heap transactions, durable log records permit table pages to be written later. Log order matters.



Checkpoint shortens recovery

With no active transactions, checkpoint flushes WAL, flushes pages, syncs the database, then truncates eligible WAL and redeclares schema/run state. Retention pins can prevent discarding consumer-needed history.

What the crash test proves

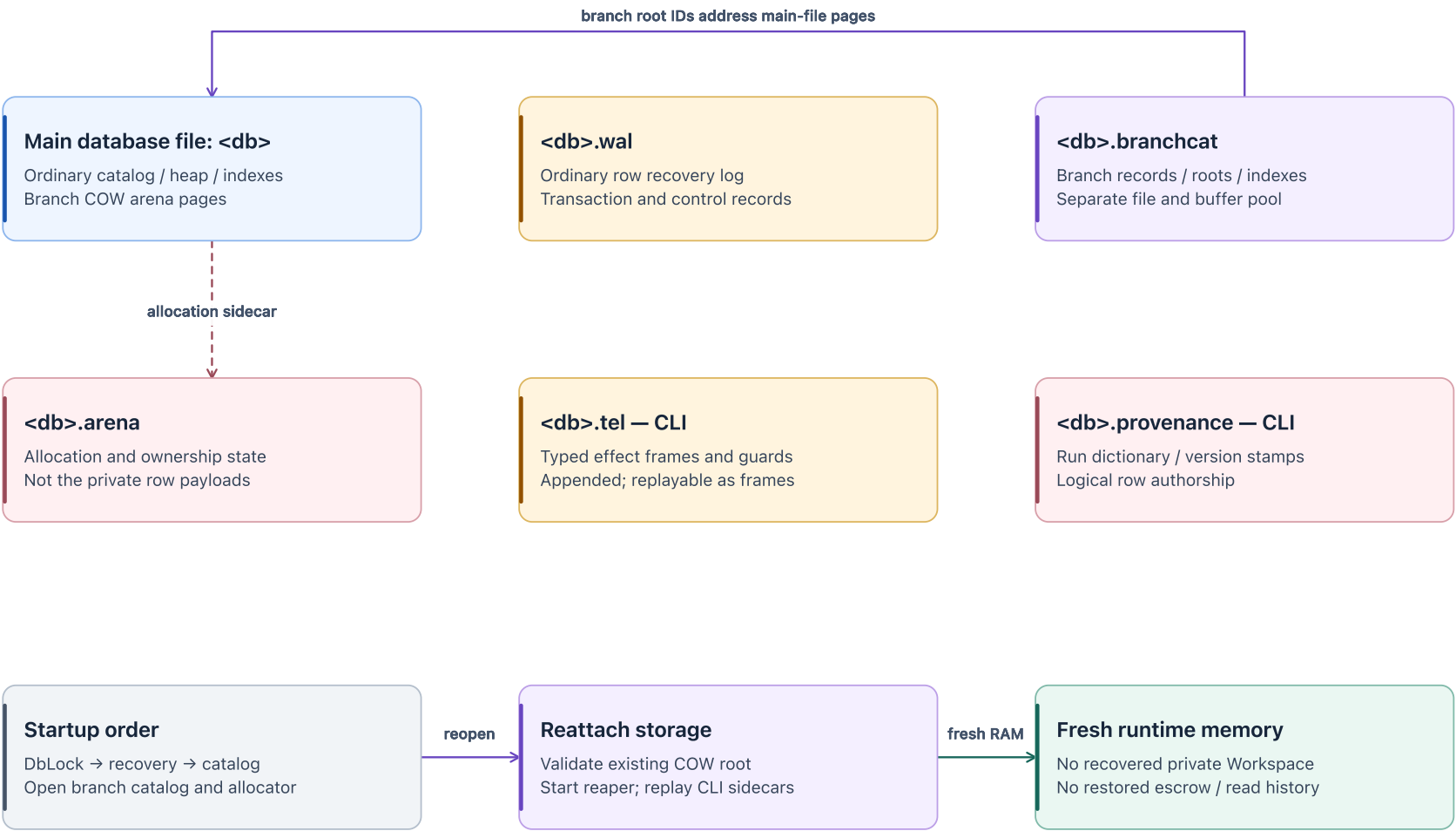
The focused merge test kills a process mid-row-publication and checks all-before or all-after rows. It does not simulate loss of the operating system page cache during machine power failure.

COW has a different local rollback tool

CowTree WriteJournal restores in-place pages after one tree-operation error and delays retirement. It is not this WAL transaction and does not make all branch sidecars crash-atomic.

What is on disk, and what reopening restores

Different files and buffer pools create different durability boundaries. Persisted components do not imply a restored agent session.



Root publication is a separate boundary

set_root writes branch metadata in the catalog pool. It is not an atomic commit across COW data, allocator state, effect log, provenance and shared heap.

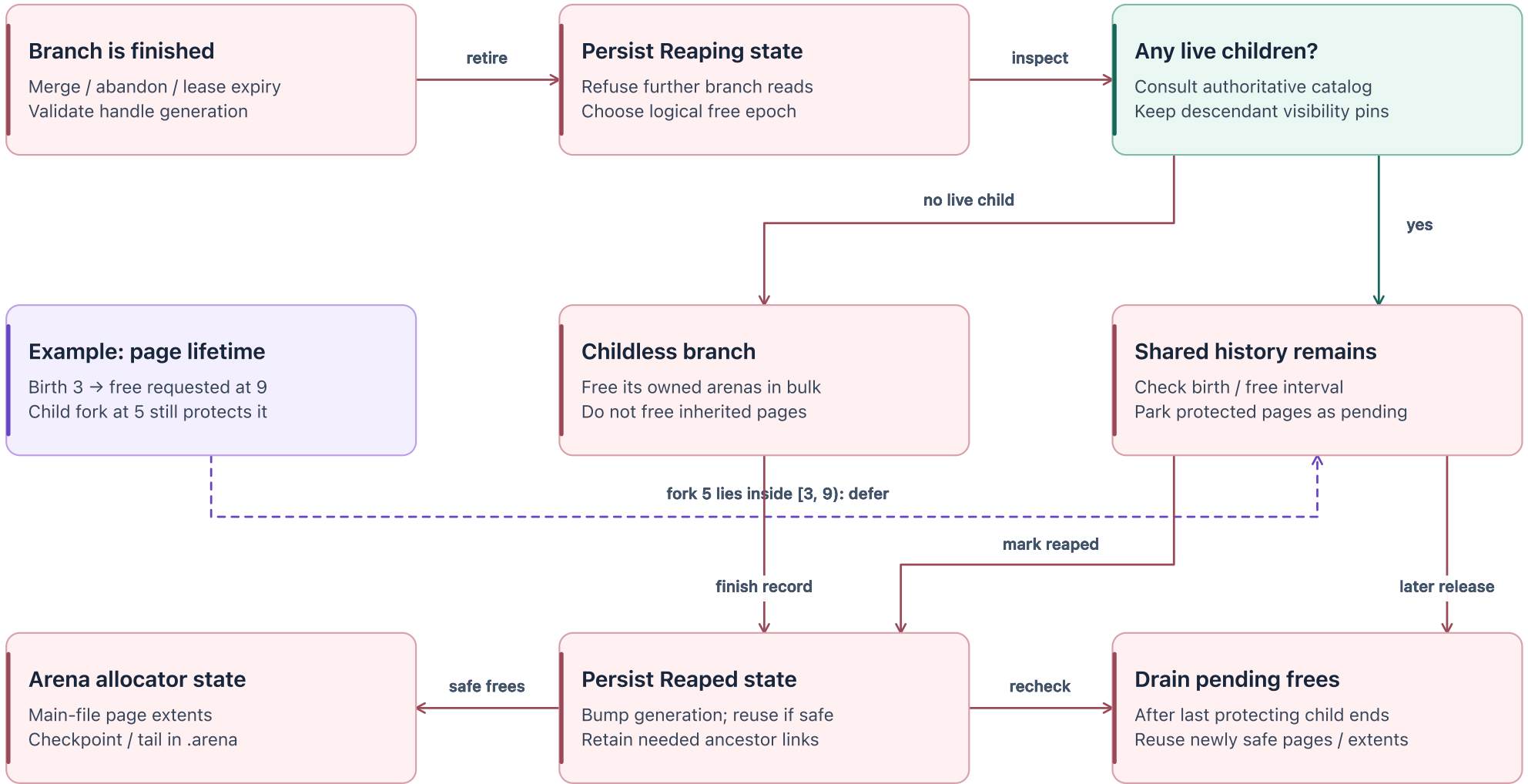
Two shipped entryptoints differ

The CLI wires durable TEL/provenance and index rebuilding after recovery. examples/pgserver.rs uses memory TEL/provenance and lacks the CLI index-rebuild block. The map describes both, not an idealized common configuration.

Component	CLI	pgserver example
Main heap / WAL / arena / branch catalog	Page/file-backed	Page/file-backed
Effect log (TEL)	DurableEffectLog → .tel	MemEffectLog
Provenance	DurableProvenanceStore → .provenance	Memory store
Index rebuild after WAL recovery	Explicitly called	No corresponding call
Unfinished agent Workspace restored	No	No

Who owns a page, and when it can be freed

The cleanup rule preserves descendants that might still reach old pages. Epoch, lease and generation are different clocks/identities.



Epoch

Logical ordering of page births, forks and frees. Also participates in deciding whether a branch-owned page is private enough to edit in place.

Lease

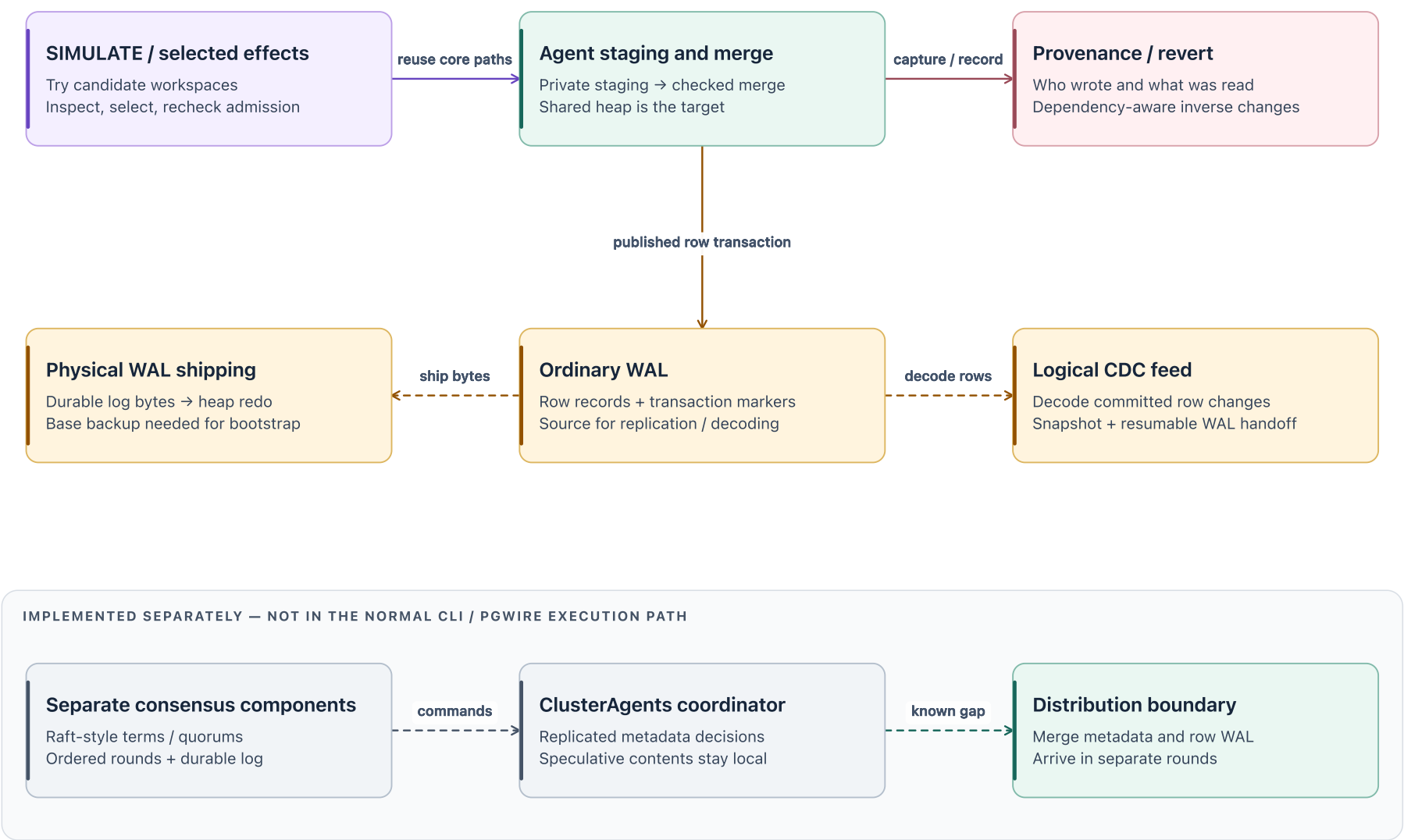
Standalone: wall-clock deadline for abandoned work; clustered components use replicated time. A background scan and cleanup do the actual reclaim; this is separate from MVCC snapshot visibility.

Generation

Identity fence for recycled branch IDs. Same numeric slot with a different generation means a different branch. Branch GC does not vacuum ordinary heap row versions.

Extensions connect to the core at specific points

These are attachments, not a second explanation of the database. Dashed arrows mean a separate/optional component connection.



Tree utilities are distinct

Page diff skips shared page IDs during paired descent. cow::merge3 is a structural base/ours/theirs merger; SQL MERGE uses operation semantics. Auxiliary content hashes do not define production page identity.

Deliberately outside the core map

Benchmark harnesses, full-text retrieval, wire-format conveniences, unwired dedup and page-delta compression utilities are not expanded. Their omission does not imply they are core execution dependencies.

Keep the mechanisms distinct.

Page

A 4 KB block addressed by a page ID. Its format depends on what it stores.

Internal / leaf

An internal tree node routes to children; a leaf holds the tree's key/value entries.

RecordId

Physical ordinary row location: page ID plus slot number.

Buffer pool

RAM cache of disk pages. Pins prevent eviction; locks protect accesses to bytes.

Snapshot

Transaction-ID boundary and active set defining visibility, not a complete copied database.

Workspace

RAM maps of private row state, before-images, effects, schema edits and related task state.

Epoch

Logical order used for sharing and reclamation. Not elapsed time or MVCC transaction identity.

TEL / effect frame

Typed operation intent and guards for a task. Separate from physical recovery WAL.

CLR

Compensation log record describing undo progress so recovery can continue after another crash.

Fingerprint

Evidence of the evaluated base, checked again before publishing; not a full serializability proof.

CDC

Committed row-change feed decoded from WAL, optionally bootstrapped with a snapshot.

Page ID

A numeric block address, not a memory pointer or a content hash.

Separator key

A boundary used to choose which child subtree can contain a search key.

Logical row ID

Identity used to match agent changes; currently derived from primary-key values.

Latch

Short-lived physical synchronization for page/tree operations; distinct from transaction visibility.

Version chain

Previous-version pointers linking a current tuple to older tuples.

Persistent map

Immutable, structurally shared map in RAM. Persistent here does not mean saved to disk.

Lease

Deadline after which background cleanup can retire abandoned branch work.

WAL

Write-ahead log: recovery information is made durable before associated dirty heap pages are written.

Guard / assertion

Guard: precondition on the target before effects. Assertion: condition on proposed results.

Quarantine

Hold an agent branch for inspection after an admission gate refuses publication.

Consensus round

Position in the replicated command order, distinct from the local WAL byte LSN.

Root page ID

The starting page for a tree. A branch stores this in its branch record.

Header / payload

The format's metadata prefix versus the node or tuple data following it.

Heap

Pages containing ordinary table tuples, organized through page directories.

MVCC

Multiple row versions plus a rule choosing which version a reader sees.

COW / shadow paging

Share existing pages; copy pages when an edit would otherwise change shared state.

Arena

A contiguous page extent owned by a branch; a branch can own multiple extents.

Generation

Version of a recycled branch identity slot, used to reject stale handles.

LSN

WAL position; page LSNs help avoid replaying the same physical update twice.

Commuting effects

Supported operations whose combination can preserve both intents; checks can still reject the result.

Checkpoint

Coordinate durable state to reduce replay history; retention rules may prevent log truncation.

Scope of this atlas

- Mapped to the source revision above. Four independent source reviews were reconciled; see review/ and verification.json for evidence and final checks.
- The atlas explains implemented mechanisms. It does not certify every combination of isolation, schema change, rollback, index access, allocation and machine power loss.
- Agent branch semantics, costs, server concurrency and restart boundaries are stated explicitly. No general branch-versus-transaction performance advantage is claimed.
- The application source was not changed. Focused existing tests support specific claims; they are not a full test-suite or exhaustive correctness proof.